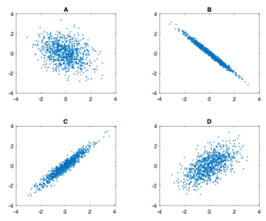
Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item →

1 / 1 point



1. This figure shows scatterplots of data (First data set is on x-axis, Second data set is on y-axis), with varying correlations. Which of these correspond to negative correlation (pick all that apply):

- [x] Plot A

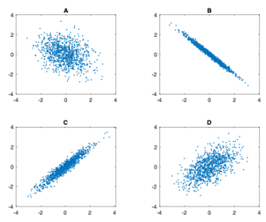
  Correct
  Correct - Negative correlation implies that as X increases, Y tends to decrease. There is not a strong correlation here, but a negative trend can be detected.

- [x] Plot B

  Correct
  Correct - Negative correlation implies that as X increases, Y tends to decrease. There is a definite negative trend observed here.

- [ ] Plot C
- [ ] Plot D

1 / 1 point



2. This figure shows scatterplots of data (First data set is on x-axis, Second data set is on y-axis), with varying correlations. Which of these correspond to negative correlation (pick all that apply):

- [x] Plot A

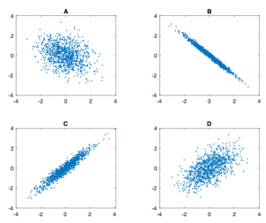
  Correct
  Correct - Negative correlation implies that as X increases, Y tends to decrease. There is not a strong correlation here, but a negative trend can be detected.

- [x] Plot B

  Correct
  Correct - Negative correlation implies that as X increases, Y tends to decrease. There is a definite negative trend observed here.

- [ ] Plot C
- [ ] Plot D

1 / 1 point



3. This figure shows scatterplots of data (First data set is on x-axis, Second data set is on y-axis), with varying correlations. Which 2 plots show the highest absolute correlation |rho|?

- [ ] Plot A
- [x] Plot B

  Correct
  Correct – A correlation of rho close to either -1 or +1 will approach a straight line. This plot does shows strong negative correlation.

- [x] Plot C

  Correct
  Correct – A correlation of rho close to either -1 or +1 will approach a straight line. This plot does shows strong positive correlation.

- [ ] Plot D

4. Given a covariance function $B(\tau) = 0.5e^{-2|\tau|}$, what is the variance of the random process? 1 / 1 point

- ( ) 0.25
- (•) 0.5
- ( ) 1
- ( ) 2

Correct
Correct! The variance is simply the covariance function evaluated at $\tau = 0$, or 0.5.

5. Given a correlation function $e^{-2|\tau|}$, what is the correlation length? 1 / 1 point

- ( ) 0.5
- (•) 1
- ( ) 2
- ( ) Undefined

Correct
Correct! The correlation length is the integral of the correlation function from -infinity to infinity. This is equivalent to the integral of $2e^{-2\tau}$ from 0 to ∞, which is $-e^{-2\tau}$ evaluated at ∞ minus $-e^{-2\tau}$ evaluated at 0. This therefore simplifies to 1.

6. Given a correlation function $1 - k * |\tau|; -1/k < \tau < 1/k$, what is the correlation length? 1 / 1 point

- ( ) 1/4k
- ( ) B: 1/2k
- ( ) 3/4k
- (•) *D: 1/k

Correct
Correct! The correlation length is the integral of the correlation function from -infinity to infinity. This is equivalent to the integral of $2(1 - k * \tau)$ from 0 to $1/k$, which is $(2\tau - k * \tau^2)$ evaluated at $1/k$ minus $(2\tau - k * \tau^2)$ evaluated at 0. This therefore simplifies to $1/k$.

7. Referring to the python notebook shared with you, which estimates the correlation function from data, experiment with different values of b (running the whole code and indicating your value of b whenever prompted). In particular, use b=10 (corresponding to a correlation length of 31.4), b=0.1 (corresponding to correlation length of 0.314) and b=0.001 (corresponding to a correlation length of 0.00314). In each case, observe what the final prediction of the correlation length is, and compare that to these known values. For example, run the entire code using b=10 at every prompt, then compare your final estimate of correlation length at the end of the code to the given correlation length of 31.4. Note that the code takes a minute to generate the data, so be patient. For which value of b is our estimate the most accurate? 1 / 1 point

- ( ) b=10
- (•) b=0.1
- ( ) b=0.001

Correct
Correct! For b=0.1, there is a balance in which the correlation length (0.314) is captured reasonably well. When b=0.001, the correlation length (.00314) is smaller than the discretization (.01), so the samples are unable to capture spatial correlations between points properly. For b=10, the correlation length (31.4) is longer than the sample length (10), so each individual sample cannot contain a full representation of the random process. Therefore, the correlation length is highly underestimated for b=10.